Ritvik Gupta

xxx-xxx-xxxx | ritvikg4@illinois.edu | linkedin.com/in/ritvikgupta11 | ritvikg.net | github.com/ritvikg4 | US Citizen

Education

University of Illinois at Urbana-Champaign GPA: 3.96/4.00
Bachelor of Science in Computer Engineering *Expected Dec 2028*

- **Relevant Coursework:** Digital Systems Laboratory (ECE 385), Analog Signal Processing (ECE 210), Data Structures (CS 225), Computer Systems and Programming (ECE 220), Discrete Math (MATH 213)

Experience

Rivian Feb 2026 – Present
Low Voltage Electronics Intern *Research Park, IL*

- Owned end-to-end PCB design for Halo test boards in Altium, informing future V1.5 and V2 hardware changes.
- Wrote EPOS motor controller firmware for Halo thermal test fixture, developed console interface for ME team use.
- Served as main POC for Halo End-of-Line testing and developed verification scripts to isolate board-level defects.
- Led critical Halo engineering investigations, performing root-cause analysis via targeted CAN bus data analysis.
- Built custom CAN front-end tool enabling real-time parameter modification for production Halo haptic tuning.

Illini Electric Motorsports - Formula SAE Sep 2025 – Present
Circuit Design Project Lead *Champaign-Urbana, IL*

- Led end-to-end development of the V2 Vehicle Control Board, integrating critical driver safety fault-management circuits and multi-node CAN communication.
- Contributed to winning 3rd place in Design and achieving the highest-judged Low Voltage score overall out of 100+ teams at FSAE Michigan 2026.
- Designed robust power delivery and analog/digital routing layout in Altium Designer, maximizing signal integrity and thermal performance under packaging constraints.
- Formulated a rigorous hardware test and validation plan, isolating fault-management corner cases to ensure vehicle reliability pre-competition.

Wolfram Research Feb 2025 – Jul 2025
Education Software Development Intern *Remote*

- Engineered full-stack web portal for High School Summer Research Program using Wolfram Cloud and JavaScript.
- Streamlined application submission, progress tracking, and program management for 100+ students and faculty.
- Designed and implemented database schema and backend logic in Wolfram Language to organize daily staff tasks.

Projects

Cellular Automata Generation of 3D Printer Infill | *Wolfram Mathematica* Sep 2024 – Mar 2025

- Implemented 3D cellular automata in Wolfram Language to algorithmically generate 3D printer infill geometries.
- Parameterized infill density and connectivity rules to optimize structural efficiency and print material usage.
- Conducted successful test prints on a Bambu Lab printer to validate automata-generated infill structures.

Parametric GVF | *Wolfram Mathematica, Java, Control Theory* Jun 2024 – Jul 2024

- Implemented a bird flocking simulation in Wolfram Language demonstrating emergent behavior from simple rules.
- Integrated a guiding vector field to direct flock behavior, achieving convergence and following of parametric paths.

Spot@MHS | *Python, Boston Dynamics* Nov 2023 – Mar 2024

- Developed comprehensive educational curriculum for the Spot robot, streamlining its use as a classroom tool.
- Designed and open-sourced a modular payload system for Spot to expand its capabilities in a cost-effective manner.

Technical Skills

Hardware: Altium Designer, CANalyzer/PCAN, Oscilloscopes, Logic Analyzers, 3D Printers (Bambu Lab)
HDL + Architecture: Verilog, Vivado, Bazel, Git
Programming Languages: Python, C, Verilog, C++, Java, JavaScript, Assembly (LC-3), Wolfram Language
Frameworks + Libraries: pytest, Flask, React, NumPy, OpenCV, PyTorch, TensorFlow, MediaPipe